



**Michigan
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Ferris Wheel Proportional Control

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L14 Team 2

Plots and Table Summary

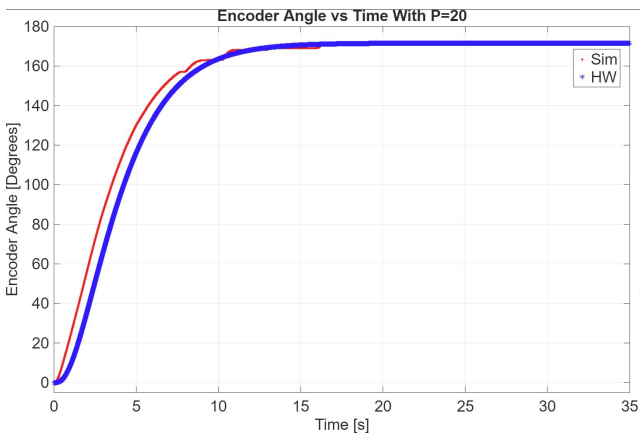


Figure 1. Plot of Simulation and Hardware Encoder Angle vs. Time with P Gain of 20 dB

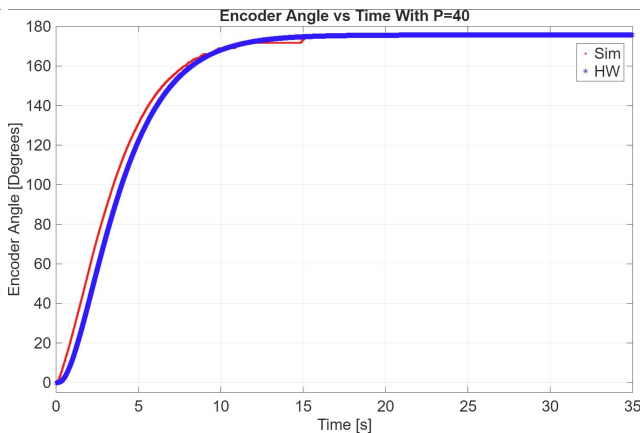


Figure 2. Plot of Simulation and Hardware Encoder Angle vs. Time with P Gain of 40 dB

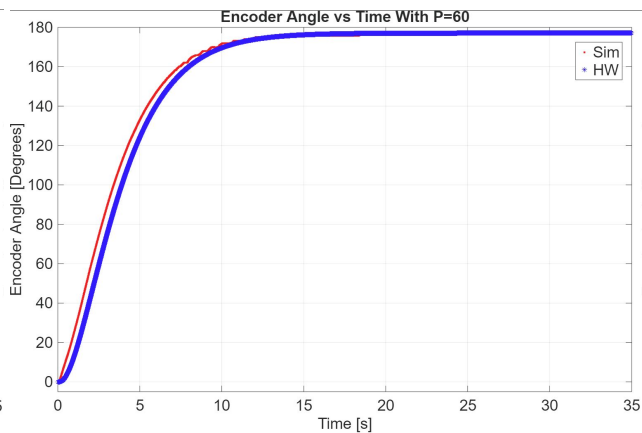


Figure 3. Plot of Simulation and Hardware Encoder Angle vs. Time with P Gain of 60 dB

Table 1. Simulation Test Observations

P Gain (dB)	Stability	Overshoot (deg)	Oscillation	Final Value (deg)	Settling Time (s)	SS Error (deg)	Peak Motor Voltage (Volts)
20	Overdamped and stable	N/A	N	171.547	14.839	8.453	2.644
40	Overdamped and stable	N/A	N	175.661	14.678	4.339	2.719
60	Overdamped and stable	N/A	N	177.076	14.717	2.924	2.742

Table 2. Hardware Test Observations

P Gain (dB)	Stability	Overshoot (deg)	Oscillation	Final Value (deg)	Settling Time (s)	SS Error (deg)	Peak Motor Voltage (Volts)
20	Overdamped and stable	N/A	N	171.072	17.54	8.928	4.096
40	Overdamped and stable	N/A	N	175.068	16.25	4.932	3.767
60	Overdamped and stable	N/A	N	177.768	19.476	2.232	4.182

Figures 1, 2, 3 present the simulation and hardware encoder angle comparison plots with a proportional gain (P) value of 20, 40, and 60 dB, respectively. Table 1 presents the simulation test observations for each of the three P gains. Green shading indicates the dynamic response characteristics which meet the engineering requirements of the Ferris Wheel. Table 2 presents the hardware test observations for each of the three P gains. The overshoot and oscillation columns for both simulation and hardware meet the engineering requirement of passenger comfort because a steady state is achieved without excessive overshoot or oscillations. The simulation settling time column meets the engineering requirement of a settling time of 16 seconds or less. The final value and steady state error for both simulation and hardware tests do not meet the engineering requirements because a final target value of 180 degrees with a range of one degree above or below is not achieved.

Dynamic Response Characterization

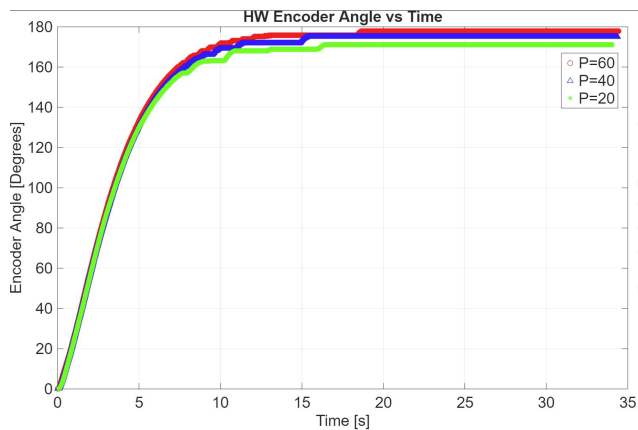


Figure 4. Plot of angular position over time for P values 20, 40, and 60 (hardware)

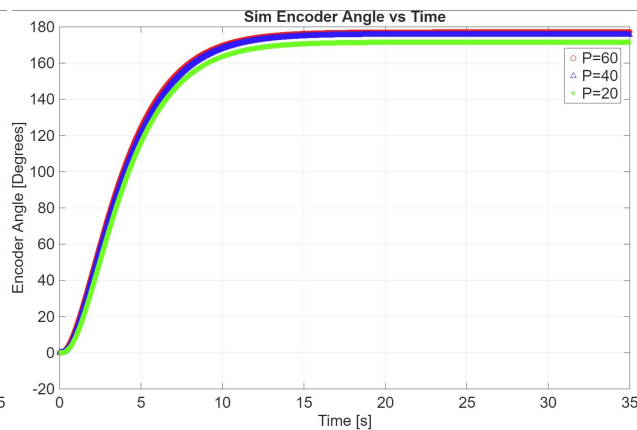


Figure 5. Plot of angular position over time for P values 20, 40 and 60 (simulation)

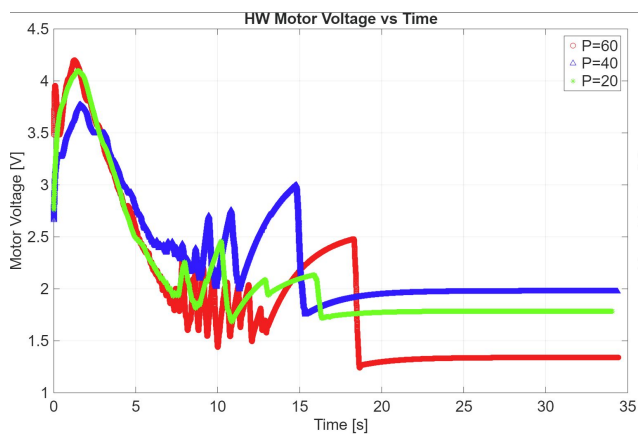


Figure 6. Plot of motor voltage over time for P values 20, 40, and 60 (hardware)

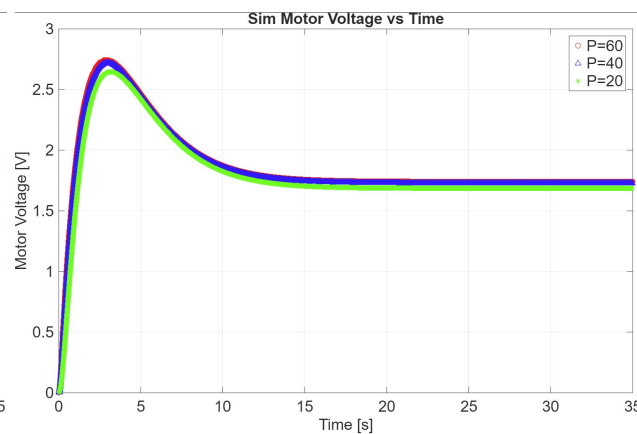


Figure 7. Plot of motor voltage over time for P values 20, 40 and 60 (simulation)

- All the measured data points remain bounded and consistent over time, making them stable responses.
- No tested values resulted in overshoot considering there were no inputs exceeding the final steady state.
- No plots involved any type of oscillation since there is no repeating, periodic back-and-forth motion.
- All the tested values settled to a settle state final value since every plot leveled out at a particular encoder angle.
- No points tested resulted in a final position within the steady-state error requirement defined in §22.1. The requirements stated that the position accuracy and stability be within one degree and none of the differences between final, stable output and input as time approaches infinity met that criteria.
- As the P values increased the steady state error decreased by more than two degrees with each P value increase for both tests. Settling time was minimally impacted in the simulation test while the settling time for the hardware test decreased by about a second before increasing by three seconds after each P value change. Increasing P values did not affect overshoot in any of the tests as seen in figures 4 and 5. Peak motor voltage was minimally impacted for the simulation as seen in figure 7, contrary to the hardware test where the voltage initially dropped by approximately 0.3 volts for the first P value increase and then increased by about 0.4 volts for the second (see figure 6).
- The hardware and simulation tests match well when it comes to stability, overshoot, and oscillation. Final values and steady state error examined minor differences with the simulation tests having slightly higher values than the hardware at P = 20 and P = 40 and the simulation having slightly lower steady state error values for each P value. Settling time and peak motor voltage proved to be marginally different between the testing, with the hardware testing consistently having two more volts than the simulation for every corresponding P value. Settling time was consistently more than two seconds longer for the hardware testing compared to the simulation as well. This type of discrepancy is most likely due to sensor error or misalignment.
- According to the data on Table 1 and Table 2, there were no P values on either test that met the engineering requirements of position accuracy and stability. All of the settling times for each P value of the simulation met the requirement while none of the hardware values did.

Impact of Proportional Gain on Controller Behavior

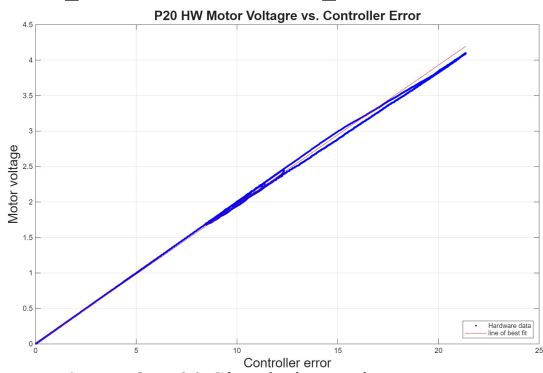


Figure 8: P20 Simulation voltage over error

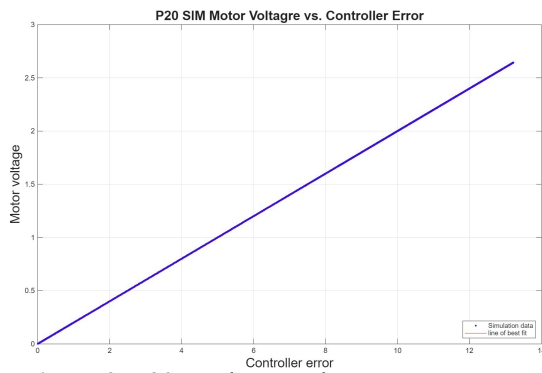


Figure 9: P20 Hardware voltage over error

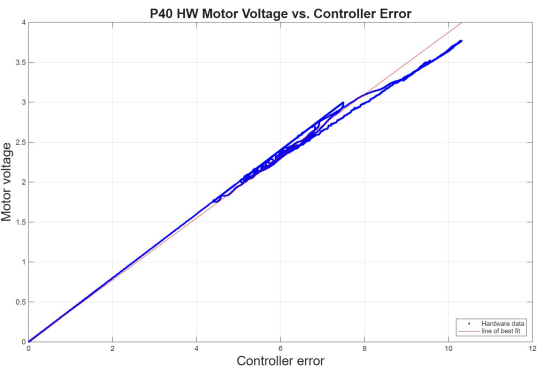


Figure 10: P40 Simulation voltage over error

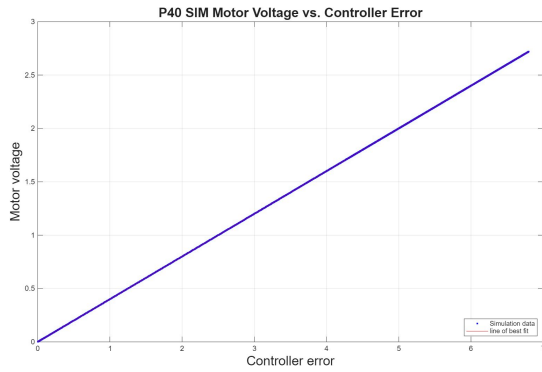


Figure 11: P40 Hardware voltage over error

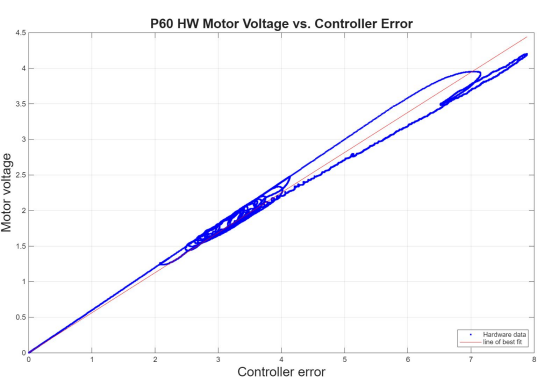


Figure 12: P60 Simulation voltage over error

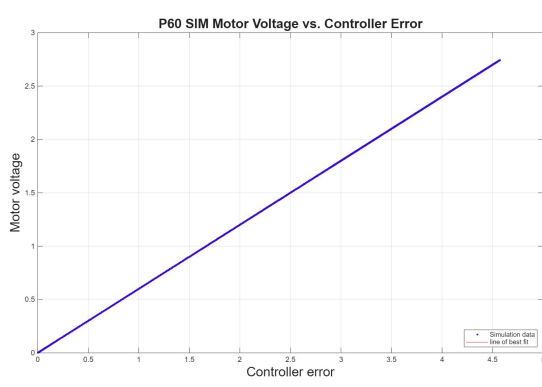


Figure 13: P60 Hardware voltage over error

Table 3: Best Fit Slope

	p20	p40	p60
Hardware	0.1964	0.3874	0.5633
Virtual	0.2	0.4	0.6

When the slopes between the simulated and hardware systems are compared it can be seen that the hardware system tends to be slightly under the systems slope. This gap only increases the higher the proportional gain is set, starting at 0.01 under for 20 ending at 0.31 under at 60.

Proportional gain facts how sever the voltage output to the motor is relative to how much error there currently is. For example a system set at a high P gain will have a larger voltage output for a smaller error distance than a low P system. Increasing a systems P will make the system reach its desired reference point.

When a system encounters a large error the voltage called for may exceed the capability of the motor. This causes the motors drivers to cap out. When this occurs the system can no longer meet the voltage output demands and will no longer linearly react to error.

When using a high proportional gain a system will tend to act aggressive and quickly, this will mainly reduce the steady state error. However, this also creates more instability in the system, increasing overshoot, and oscillation frequency.